Distance geometry and geometric algebra

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Distance Geometry and Geometric Algebra

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Abstract

> ... Möbius's invention of homogeneous coordinates was one of the most far-reaching ideas in the history of mathematics: comparable to Leibnitz's invention of differentials ...
>
> *H.S.M. Coxeter, Introduction to Geometry, pg. 221*

> ... the inhabitants of a hyperbolic world would also study horospherical geometry, which is the same as Euclidean geometry!
>
> *H.S.M. Coxeter, Introduction to Geometry, pg. 304*

As part of his program to unify linear algebra and geometry using the language of Clifford algebra, David Hestenes has constructed an isomorphism between the conformal group and the orthogonal group of a space two dimensions higher, thus obtaining homogeneous coordinates for conformal geometry[1]. In this paper we show that his construction is the Clifford algebra analogue of a hyperbolic model of Euclidean geometry that has actually been known since Bolyai, Lobachevsky and Gauss, and we explore its wider invariant theoretic implications. In particular, we show that the Euclidean distance function has a very simple representation in this model, as demonstrated by J. J. Seidel[18].

1 Historical Introduction

In a fascinating recent paper, the *Design of Linear Algebra and Geometry*[1], David Hestenes has shown how a well-known representation[2, 3] of the conformal group of $\mathbf{R}^{p,q}$ in $\mathbf{R}^{p+1,q+1}$ can be elegantly constructed using the language of Clifford algebra, or *geometric algebra* as Hestenes prefers to call it[4]. Hestenes calls his construction the *conformal split*.

Since this representation is independent of scale, the vectors in $\mathbf{R}^{p+1,q+1}$ can be thought of as homogeneous coordinates for conformal geometry. Although this is not the only means of representing conformal transformations using Clifford algebra[5, 6, 7, 8], the conformal split representation permits a remarkably simple analytic description of the distance function, and has interesting connections with hyperbolic geometry (see below). The purpose of this paper is to show how the classical work by Menger, Blumenthal and Seidel on *distance geometry* can be interpreted in the context of this representation, and to consider its wider invariant theoretic consequences.

Before we begin, however, it is worth taking a few pages to explain *why* this work is important, and to place it in its proper historical perspective. Our story begins, like so much else in this area, with Hermann Günther Grassmann (1809 – 1877). Although his contributions have finally begun to receive the recognition they have long deserved, many aspects of his work remain unappreciated. One of these aspects stems from the fact that, for Grassmann, the primitive geometrical object was almost always points instead of vectors. Alternatively, one could say that he regarded the notion of position as being more fundamental than the notion of direction (although they can of course be defined in terms of one another).

Thus, in the *Ausdehnungslehre* of 1844, Grassmann first defined the outer product for pairs of points, and he regarded it as a *line-bound vector* (also known as a *sliding vector* in mechanics) rather than as an areal magnitude[9, 10]. His definition of the inner product, which came later in a work known as *Die Geometrische Analyse* (1847), was originally given for vectors. In the last section of that work, however, he defined the inner product of a pair of points $\mathbf{p} = (\mathbf{r}+\mathbf{v})/2$, $\mathbf{q} = (\mathbf{r}-\mathbf{v})/2$ as $\mathbf{p}\cdot\mathbf{q} = \mathbf{r}^2/4 - \mathbf{v}^2/4$, where $\mathbf{v} = \mathbf{p}-\mathbf{q}$, $\mathbf{r} = \mathbf{p}+\mathbf{q}$ and the square of the midpoint $\mathbf{r}^2/4$ was an undefined *inner magnitude*. Setting $\mathbf{r}^2 = -\mathbf{v}^2$ yields the inner product $\mathbf{p}\cdot\mathbf{q} = -\frac{1}{2}(d(\mathbf{p},\mathbf{q}))^2$, where $d(\mathbf{p},\mathbf{q})$ is the distance from $\mathbf{p}$ to $\mathbf{q}$.

The many other contributions of Grassmann to geometric algebra have been nicely summarized by David Hestenes in the first chapter of his book *New Foundations for Classical Mechanics*[11]. We now turn to the work of Lobachevsky, Bolyai and Gauss on non-Euclidean geometry. These early investigators observed that a certain type of surface in hyperbolic geometry known as the *horosphere* is metrically equivalent to a Euclidean space. A horosphere is the hyperbolic analogue of a parabola, in that it is a quadratic surface containing one infinite point (i.e. a sphere of infinite radius). By metrically equivalent, we mean that there exists a bijective mapping between the horosphere and a Euclidean space such that lengths of the geodesics between pairs of points in the horosphere (horocycles) are equal to the Euclidean distances between their images under this map. A student of Gauss named F. A. Wachter (1792 – 1817) went on to show that horospheres in fact constitute a *non-Euclidean model of Euclidean geometry*.

To see how this works, we recall that Poincaré's model of an n-dimensional hyperbolic space consists of the upper half space $\{\mathbf{x} \in \mathbf{R}^n \mid x_n > 0\}$ together with the Riemannian metric $((d\,x_1)^2 + \cdots + (d\,x_n)^2)/x_n^2$ (see e.g. the text by

Thorpe[12]). In this model, the horospheres are easily seen to be spheres that are tangent the plane $x_n = 0$, together with the planes $x_n = c$ ($c > 0$). Since all horospheres are metrically equivalent up to rescaling (exercise!), it is sufficient to consider only the latter type of horosphere, on which the Poincaré metric is obviously equal to the usual metric up to a scale factor. A little thought will also show that since the entire plane at infinity $x_n = 0$ is the limit of a pencil of concentric horospheres, this plane is also a Euclidean space!

Another important branch of our story reaches back to the work of Arthur Cayley, who published a paper in 1841 entitled *A Theorem in the Geometry of Position*, in which he showed that the squared volume of a simplex $\mathbf{p}_0, \ldots, \mathbf{p}_n$ could be expressed in terms of the following determinant:

$$
\begin{aligned}
&D(\mathbf{p}_0, \ldots, \mathbf{p}_n) \\
&:= \; -\left(\frac{1}{n!}\right)^2 \begin{vmatrix}
0 & 1 & 1 & \cdots & 1 \\
1 & 0 & -\frac{1}{2}\, d(\mathbf{p}_0, \mathbf{p}_1)^2 & \cdots & -\frac{1}{2}\, d(\mathbf{p}_0, \mathbf{p}_n)^2 \\
1 & -\frac{1}{2}\, d(\mathbf{p}_0, \mathbf{p}_1)^2 & 0 & \cdots & -\frac{1}{2}\, d(\mathbf{p}_1, \mathbf{p}_n)^2 \\
\vdots & \vdots & \vdots & \ddots & \vdots \\
1 & -\frac{1}{2}\, d(\mathbf{p}_0, \mathbf{p}_n)^2 & -\frac{1}{2}\, d(\mathbf{p}_1, \mathbf{p}_n)^2 & \cdots & 0
\end{vmatrix} .
\end{aligned}
$$

Thus this expression is always nonnegative and zero only if the points are affinely dependent. The determinant itself is called a *Cayley-Menger determinant*, while the corresponding matrix will be referred to here as a Cayley-Menger matrix.

Almost a century later, Karl Menger proved that the nonnegativity of these expressions is in fact a sufficient condition for any symmetric matrix of nonnegative real numbers with zeros down the diagonal to be the matrix of squared distances among a set of points in Euclidean space[13]. He then went on to demonstrate that this fact could be used to derive a new system of axioms for Euclidean geometry whose primitive notion was the interpoint distance[14]. This work was later expanded to non-Euclidean geometries by Seidel[15] and Blumenthal[16, 17]. In particular, Seidel[18] showed how Wachter's non-Euclidean model of Euclidean geometry enables us to interpret Cayley-Menger determinants as Gramians in a Lorentz space.

The final conceptual stream feeding into this paper comes from *invariant theory*[19, 20, 21]. The objects of interest here are essentially the representations of transformation groups that are induced by their actions on functions of the coordinates, and in particular the trivial representations that arise when the function is an invariant of the group action. It is obvious that the ratios of the determinants of the homogeneous coordinates of points in a projective space are invariants of the projective group, and hence the determinants themselves are known as *relative invariants*. The first fundamental theorem of invariant theory states that in fact all relative invariants of the projective group are but functions of these determinants (called *brackets*), and for many years the search for similar theorems describing the invariants of other group actions was a major area of mathematics. This era came to an abrupt end when David Hilbert found a nonconstructive proof of the existence of a finite algebraic basis for the invariants of any such group action[22, 23].

In recent years invariant theory has experienced a remarkable renaissance, largely because it has been found that *constructive* derivations using these methods are a powerful means of formulating and solving concrete geometric problems. Most of this work has concentrated on projective geometry, where the Grassmann or (a little more generally) Clifford algebra[24] associated with a vector space is used to formulate the problem, which may then be translated into a question concerning the existence of solutions to polynomial equations in the brackets and ultimately, the coordinates themselves[25, 26, 27]. These solutions may then be found mechanically by means of such computer algebra methods as Gröbner bases[28, 29].

It is somewhat surprising that similar methods have not been developed for the geometry that is of greatest applied interest: Euclidean geometry. Indeed, it has only recently been demonstrated that it is possible to give invariant-theoretic proofs of theorems in Euclidean geometry using the distances as coordinates[30], and a proof of the first fundamental theorem for Euclidean geometry, namely that the interpoint distances squared constitute a complete system of invariants for the Euclidean group, has only recently been fully worked out[31]. Part of the reason that the Euclidean case has not been developed as highly is the lack of a suitable algebraic system wherein the geometric problems can be concisely stated, and then automatically translated into a question regarding the solvability of a system of polynomials that are invariant under a simple action of the Euclidean group. The combination of Clifford algebra with the homogeneous coordinates for conformal (and hence Euclidean) geometry obtained via Hestenes' conformal split may provide us with that missing link.

2 The Basic Idea

We begin by deriving Cayley and Menger's results using the language of matrix algebra[32, 33]. This language has the advantage of being familiar to all our potential readers, but the disadvantage of obscuring the geometric meaning of the derivation. In the next section, we will show how the language of geometric algebra makes it easy to interpret the derivation geometrically, thus demonstrating the advantages of geometric algebra.

We begin by recalling a few well-known definitions and facts from matrix algebra. The *Gramian* of a set of vectors $\{\mathbf{v}_1, \ldots, \mathbf{v}_n\}$ in a Euclidean vector space is the determinant $G(\mathbf{v}_1, \ldots, \mathbf{v}_n) = \det(\mathbf{M})$ of their matrix of inner products:

$$\mathbf{M} \;=\; \begin{pmatrix} \mathbf{v}_1 \cdot \mathbf{v}_1 & \cdots & \mathbf{v}_1 \cdot \mathbf{v}_n \\ \vdots & \ddots & \vdots \\ \mathbf{v}_n \cdot \mathbf{v}_1 & \cdots & \mathbf{v}_n \cdot \mathbf{v}_n \end{pmatrix} .$$

Since the above matrix $\mathbf{M}$, called the *metric matrix*, is positive semidefinite of rank equal to the dimension of the subspace spanned by the vectors $\mathbf{v}_1, \ldots, \mathbf{v}_n$, the Gramian is nonnegative. Moreover, given any such matrix $\mathbf{M}$, we may always find an $n \times r$ square root $\mathbf{R}\,\mathbf{R}^\top = \mathbf{M}$, where $r = \mathrm{rank}(\mathbf{M})$ and the rows of $\mathbf{R}$ are coordinates for the vectors $\mathbf{v}$. Since $\det(\mathbf{M}) = \det(\mathbf{R})^2$ when $r = n$, the Gramian is in fact just the squared volume of the parallelepiped spanned by the vectors.

Theorem (Menger[13]): A matrix of nonnegative real numbers with zeros down the diagonal is a matrix of squared distances among a set of points in an n-dimensional Euclidean space if and only if the Cayley-Menger determinants formed from all possible $k \times k$ principle submatrices are always nonnegative and vanish whenever $k > n + 1$.

PROOF: Given points $\mathbf{p}_0, \ldots, \mathbf{p}_n$, let $\mathbf{v}_i = \mathbf{p}_i - \mathbf{p}_0$ for $i = 1, \ldots, n$, so that by the law of cosines

$$\mathbf{v}_i \cdot \mathbf{v}_j = \tfrac{1}{2}\left((d(\mathbf{p}_0, \mathbf{p}_i))^2 + (d(\mathbf{p}_0, \mathbf{p}_j))^2 - (d(\mathbf{p}_i, \mathbf{p}_j))^2\right) \ .$$

Then, since the value of a determinant is invariant under elementary row / column operations, on making this substitution we find that: $G(\mathbf{v}_1, \ldots, \mathbf{v}_n)$

$$= \begin{vmatrix} 1 & 0 & 0 & \ldots & 0 \\ 0 & 1 & 0 & \ldots & 0 \\ 0 & 0 & \mathbf{v}_1 \cdot \mathbf{v}_1 & \ldots & \mathbf{v}_1 \cdot \mathbf{v}_n \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \mathbf{v}_n \cdot \mathbf{v}_1 & \ldots & \mathbf{v}_n \cdot \mathbf{v}_n \end{vmatrix}$$

$$= -\begin{vmatrix} 0 & 1 & 1 & \ldots & 1 \\ 1 & 0 & 0 & \ldots & 0 \\ 1 & 0 & \mathbf{v}_1 \cdot \mathbf{v}_1 & \ldots & \mathbf{v}_1 \cdot \mathbf{v}_n \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & 0 & \mathbf{v}_n \cdot \mathbf{v}_1 & \ldots & \mathbf{v}_n \cdot \mathbf{v}_n \end{vmatrix}$$

$$= -\begin{vmatrix} 0 & 1 & 1 & \ldots & 1 \\ 1 & 0 & -\frac{1}{2}d(\mathbf{p}_0, \mathbf{p}_1)^2 & \ldots & -\frac{1}{2}d(\mathbf{p}_0, \mathbf{p}_n)^2 \\ 1 & -\frac{1}{2}d(\mathbf{p}_0, \mathbf{p}_1)^2 & 0 & \ldots & -\frac{1}{2}d(\mathbf{p}_1, \mathbf{p}_n)^2 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & -\frac{1}{2}d(\mathbf{p}_0, \mathbf{p}_n)^2 & -\frac{1}{2}d(\mathbf{p}_1, \mathbf{p}_n)^2 & \ldots & 0 \end{vmatrix} .$$

Up to a factor of $(n!)^{-2}$, the last determinant above is just the Cayley-Menger determinant $D(\mathbf{p}_0, \ldots, \mathbf{p}_n)$. Since the elements of this determinant are independent of the location of the origin, so are all of the above determinants.

Combined with the geometric interpretation of the Gramian as the squared volume of the parallelepiped spanned by the constituent vectors, this derivation gives us Cayley's result. In addition, by performing the same derivation in reverse and taking a suitable square root of the metric matrix, we obtain coordinates for a set of points in Euclidean space of dimension two less than the rank of the Cayley-Menger matrix, whose mutual distances are equal to those in the Cayley-Menger matrix. This proves Menger's characterization of Euclidean distances.

Remarks: The very first step of the above derivation, in which we augmented the metric matrix by two additional rows and columns, can be viewed geometrically as embedding the vectors in a vector space two dimensions higher; the following swap of the first two rows then changes the signature to a Lorentz space $\mathbf{R}^{n+1,1}$. Thus, as shown by Seidel[18], the Cayley-Menger determinant is essentially the Gramian of a set of vectors in an $(n+2)$-dimensional Lorentz space. In the next section we show that these vectors are the same as those which occur in the homogeneous parametrization of conformal geometry used by Hestenes[1].

We note that the three-point Cayley-Menger determinant factorizes as follows: $D(\mathbf{p}_0, \mathbf{p}_1, \mathbf{p}_2)$

$$= \tfrac{1}{16}(d_{01} + d_{02} + d_{12})(d_{01} + d_{02} - d_{12})(d_{01} - d_{02} + d_{12})(-d_{01} + d_{02} + d_{12})$$

where the distances $d_{01} = |\mathbf{p}_0 - \mathbf{p}_1|$, etc. From this it is easily seen that the nonnegativity of this expression is equivalent to all three triangle inequalities among the distances. Unfortunately, the four and higher-point expressions do not factorize, and thus are both algebraically and geometrically much more complicated[32, 33].

3 The Conformal Split

Now suppose we are given a nondegenerate metric vector space $\mathbf{R}^{n+1,1}$ of signature $n+1$ (positive) and 1 (negative), and let $\mathbf{e}_{-1}, \mathbf{e}_0, \ldots, \mathbf{e}_n$ be the canonical basis of $\mathbf{R}^{n+1,1}$ with $\mathbf{e}_{-1}^2 = -1$, $\mathbf{e}_i^2 = 1$ for $i = 0, \ldots, n$, and $\mathbf{e}_i \cdot \mathbf{e}_j = 0$ for all $i \neq j$. If juxtaposition denotes the Clifford product associated with $\mathbf{R}^{n+1,1}$ by its quadratic form $\mathbf{x}^2 = -x_{-1}^2 + x_0^2 + \ldots + x_n^2$, we define the bivector

$$\mathbf{F} := \mathbf{e}_{-1}\,\mathbf{e}_0 = \mathbf{e}_{-1} \wedge \mathbf{e}_0 = -\mathbf{e}_0\,\mathbf{e}_{-1}$$

along with vectors $\mathbf{f}_\oplus$, $\mathbf{f}_\ominus$

$$\mathbf{f}_{\oplus} := \tfrac{1}{2}(\mathbf{e}_0 \pm \mathbf{e}_{-1}) = \tfrac{1}{2}(1 \pm \mathbf{F})\,\mathbf{e}_0$$

so that

$$\mathbf{f}_{\oplus}^2 = 0, \quad \mathbf{f}_\oplus \cdot \mathbf{f}_\ominus = \tfrac{1}{2} \quad \text{and} \quad \mathbf{f}_\oplus \wedge \mathbf{f}_\ominus = \tfrac{1}{2}\mathbf{F}\ .$$

Here, the "$\wedge$" refers to the outer product canonically associated with the Clifford algebra $C\ell_{n+1,1}$, as shown by M. Riesz (cf. Lounesto, this volume).

We now let $\mathcal{N} := \{\mathbf{v} \in \mathbf{R}^{n+1,1} \mid \mathbf{v}^2 = 0\}$ be the null or light cone in $\mathbf{R}^{n+1,1}$, and we let $\breve{\mathcal{N}}$ be the "parabola" $\{\mathbf{v} \in \mathcal{N} \mid \mathbf{v} \cdot \mathbf{f}_\oplus = -\frac{1}{2}\}$ given by the intersection of the null cone and the hyperplane $\mathbf{v} \cdot \mathbf{f}_\oplus = -\frac{1}{2}$. Then left-multiplication by the bivector $\mathbf{F}$ defines a linear injection from $\breve{\mathcal{N}}$ into $C\ell_{n+1,1}$, which Hestenes calls the *conformal split*:[34]

$$\begin{aligned} \mathbf{F}\,\mathbf{x} &= \mathbf{F} \cdot \mathbf{x} + \mathbf{F} \wedge \mathbf{x} \\ &= \mathbf{f}_\ominus + \mathbf{X}^2 \mathbf{f}_\oplus + \mathbf{X}\ , \end{aligned}$$

where $\mathbf{x} \in \breve{\mathcal{N}}$, $\mathbf{X} := \mathbf{F} \wedge \mathbf{x} = \mathbf{x} \wedge \mathbf{F}$, and $\mathbf{X}^2 = 2\,\mathbf{x} \cdot \mathbf{f}_\ominus \in \mathbf{R}$. Since $\mathbf{F}^2 = 1$,

$\mathbf{F}\cdot\mathbf{x} = -\mathbf{x}\cdot\mathbf{F}$ and $\mathbf{f}_\oplus\mathbf{X} = \mathbf{X}\,\mathbf{f}_\oplus$, it follows that for all $\mathbf{x},\mathbf{y} \in \breve{\mathcal{N}}$:

$$\begin{aligned}
\mathbf{x}\,\mathbf{y} &= (\mathbf{x}\,\mathbf{F})\,(\mathbf{F}\,\mathbf{y}) \\
&= (-\mathbf{f}_\ominus - \mathbf{X}^2\mathbf{f}_\oplus + \mathbf{X})\,(\mathbf{f}_\ominus + \mathbf{Y}^2\mathbf{f}_\oplus + \mathbf{Y}) \\
&= \mathbf{X}\,\mathbf{Y} - (\mathbf{X}^2\,\mathbf{f}_\oplus\,\mathbf{f}_\ominus + \mathbf{Y}^2\,\mathbf{f}_\ominus\,\mathbf{f}_\oplus) \\
&\quad + (\mathbf{Y}^2\,\mathbf{X} - \mathbf{Y}\,\mathbf{X}^2)\,\mathbf{f}_\oplus + (\mathbf{X}-\mathbf{Y})\,\mathbf{f}_\ominus \\
&= \langle\mathbf{X}\,\mathbf{Y}\rangle_2 + \langle\mathbf{X}\,\mathbf{Y}\rangle_0 - \tfrac{1}{2}(\mathbf{X}^2(1+\mathbf{F}) + \mathbf{Y}^2(1-\mathbf{F})) \\
&\quad + \mathbf{X}(\mathbf{Y}-\mathbf{X})\mathbf{Y}\,\mathbf{f}_\oplus + (\mathbf{X}-\mathbf{Y})\,\mathbf{f}_\ominus \\
&= \mathbf{X}\underline{\triangle}\mathbf{Y} - \tfrac{1}{2}D(\mathbf{X},\mathbf{Y}) + \tfrac{1}{2}(\mathbf{Y}^2 - \mathbf{X}^2)\mathbf{F} \\
&\quad + \mathbf{X}(\mathbf{Y}-\mathbf{X})\mathbf{Y}\,\mathbf{f}_\oplus + (\mathbf{X}-\mathbf{Y})\,\mathbf{f}_\ominus
\end{aligned}$$

Here $D(\mathbf{X},\mathbf{Y}) := \mathbf{X}^2 + \mathbf{Y}^2 - 2\langle\mathbf{X}\,\mathbf{Y}\rangle_0$ is $-2\langle\mathbf{x}\,\mathbf{y}\rangle_0 = -2\,\mathbf{x}\cdot\mathbf{y}$, and $\mathbf{X}\underline{\triangle}\mathbf{Y} := \langle\mathbf{X}\,\mathbf{Y}\rangle_2 = \mathbf{X}\,\mathbf{Y} - \langle\mathbf{X}\,\mathbf{Y}\rangle_0 = (\mathbf{x}\wedge\mathbf{y}\wedge\mathbf{F})/\mathbf{F}$ is the bivector part of $\mathbf{X}\,\mathbf{Y}$.

It should be noted that although "$\underline{\triangle}$" is an associative and alternating product, it is *not* the outer product in $C\ell_{n+1,1}$ (which would be zero, because $(\mathbf{F}\wedge\mathbf{x})\wedge(\mathbf{F}\wedge\mathbf{y}) = 0$). Moreover, since

$$|\,\mathbf{x}\wedge\mathbf{y}\,|^2 \;=\; (\mathbf{x}\cdot\mathbf{y})^2 - \mathbf{x}^2\mathbf{y}^2 \;=\; (\mathbf{x}\cdot\mathbf{y})^2\ ,$$

where $\mathbf{x}^2 = \mathbf{y}^2 = 0$, we have also $|\,\mathbf{x}\wedge\mathbf{y}\,| = \pm\frac{1}{2}\,D(\mathbf{X},\mathbf{Y})$.

For future reference, we record a few results concerning the geometry of the "parabola" $\breve{\mathcal{N}}$:

Lemma: Given any $\mathbf{z}\in\mathbf{R}^{n+1,1}$ with $\mathbf{z}\cdot\mathbf{f}_\oplus = -\frac{1}{2}$, the intersection of the plane spanned by $\mathbf{z}$ and $\mathbf{f}_\oplus$ with $\breve{\mathcal{N}}$ is given by $\breve{\mathbf{z}} := \mathbf{z} + \mathbf{z}^2\mathbf{f}_\oplus$. In particular, if

$\mathbf{z}_i = \mathbf{y}_i - \mathbf{f}_\ominus$ for $i = 1, \ldots, k$ with $\mathbf{y}_i \in \mathbf{F}^\perp := \{\mathbf{x} \in \mathbf{R}^{n+1,1} \mid \mathbf{x} \cdot \mathbf{F} = 0\}$, then $\breve{\mathbf{z}}_i = \mathbf{y}_i - \mathbf{f}_\ominus + \mathbf{y}_i^2 \mathbf{f}_\oplus \in \breve{\mathcal{N}}$, and for any coefficients $\omega_1, \ldots, \omega_k$ with $\omega_1 + \cdots + \omega_k = 0$ we have $\Sigma_i\, \omega_i\, \mathbf{z}_i := \Sigma_i\, \omega_i\, (\mathbf{y}_i - \mathbf{f}_\ominus) = \Sigma_i\, \omega_i\, \mathbf{y}_i = 0$ if and only if $\breve{\mathbf{z}}_1, \ldots, \breve{\mathbf{z}}_k$, $\mathbf{f}_\oplus$ are linearly dependent.

Proof: The first assertion is immediate since $\breve{\mathbf{z}} \cdot \mathbf{f}_\oplus = -\frac{1}{2}$ and $\breve{\mathbf{z}}^2 = \mathbf{z}^2 + 2\,\mathbf{z}^2\,\mathbf{z} \cdot \mathbf{f}_\oplus = 0$. To prove the second, we first observe that if $\Sigma_i\, \omega_i\, \mathbf{y}_i = 0$ with $\omega_1 + \cdots + \omega_k = 0$, then $\Sigma_i\, \omega_i\, \breve{\mathbf{z}}_i = \mathbf{f}_\oplus\, \Sigma_i\, \omega_i\, \mathbf{z}_i^2$ so that $\breve{\mathbf{z}}_1, \ldots, \breve{\mathbf{z}}_k$ and $\mathbf{f}_\oplus$ are dependent, as desired. Conversely, if $\Sigma_i\, \omega_i\, \breve{\mathbf{z}}_i = \mathbf{f}_\oplus$, then $\Sigma_i\, \omega_i\, \breve{\mathbf{z}}_i \cdot \mathbf{f}_\oplus = \Sigma_i\, \omega_i\, \mathbf{f}_\ominus \cdot \mathbf{f}_\oplus = 0$ so that $\omega_1 + \cdots + \omega_k = 0$. Thus $\Sigma_i\, \omega_i\, \breve{\mathbf{z}}_i \cdot \mathbf{f}_\ominus = \frac{1}{2} \Sigma_i\, \omega_i\, \mathbf{y}_i^2 = \frac{1}{2}$ and hence $\Sigma_i\, \omega_i\, \mathbf{y}_i = 0$ as desired. ▫

Note that for $\mathbf{z} = \mathbf{y} - \mathbf{f}_\ominus$, $\mathbf{F}\, \breve{\mathbf{z}} = \mathbf{Y} + \mathbf{f}_\ominus + \mathbf{Y}^2 \mathbf{f}_\oplus$.

We now recall that the inner product of two simple k-vectors $\mathbf{A} = \mathbf{a}_1 \wedge \cdots \wedge \mathbf{a}_k$ and $\mathbf{B} = \mathbf{b}_1 \wedge \cdots \wedge \mathbf{b}_k$ can be expanded as the determinant $\det(\mathbf{a}_i \cdot \mathbf{b}_j)$.

Theorem: Given vectors $\mathbf{x}_0, \ldots, \mathbf{x}_k \in \breve{\mathcal{N}}$, the quantities $D(\mathbf{X}_i, \mathbf{X}_j) = -2\, \mathbf{x}_i \cdot \mathbf{x}_j$ $(i, j = 0, \ldots, k)$ constitute the squared distances among a set of points in a Euclidean space of dimension at most n. There exist sets of vectors on the parabola $\breve{\mathcal{N}}$ such that the dimension of this Euclidean space is exactly n.

Proof: Applying the above determinant expansion to $|\; 2\, \mathbf{f}_\oplus \wedge \mathbf{x} \wedge \mathbf{y} \;|^2$, we obtain:

$$
\begin{aligned}
& 4\, (\mathbf{f}_\oplus \wedge \mathbf{x} \wedge \mathbf{y})\, (\mathbf{y} \wedge \mathbf{x} \wedge \mathbf{f}_\oplus) \\
= \quad & 4 \begin{vmatrix} \mathbf{f}_\oplus^2 & \mathbf{x} \cdot \mathbf{f}_\oplus & \mathbf{y} \cdot \mathbf{f}_\oplus \\ \mathbf{x} \cdot \mathbf{f}_\oplus & \mathbf{x}^2 & \mathbf{x} \cdot \mathbf{y} \\ \mathbf{y} \cdot \mathbf{f}_\oplus & \mathbf{x} \cdot \mathbf{y} & \mathbf{y}^2 \end{vmatrix}
\end{aligned}
$$

$$
\begin{aligned}
&= \begin{vmatrix} 0 & 1 & 1 \\ 1 & 0 & -\frac{1}{2}\,D(\mathbf{X},\mathbf{Y}) \\ 1 & -\frac{1}{2}\,D(\mathbf{X},\mathbf{Y}) & 0 \end{vmatrix} \\
&= \quad -D(\mathbf{X},\mathbf{Y})\ .
\end{aligned}
$$

This obviously generalizes to arbitrary sets of vectors $\mathbf{x}_0\,,\ldots,\mathbf{x}_k \in \breve{\mathcal{N}}$, thus showing that Cayley-Menger determinants are $-(1/k!)^2$ times the Gramians of vectors on the parabola $\breve{\mathcal{N}}$ together with $2\,\mathbf{f}_\oplus$. The only maximal degenerate subspaces of $\mathbf{R}^{n+1,1}$ are the hyperplanes tangent to its null cone $\mathcal{N}$, and since $|\mathbf{x}_i \wedge \mathbf{f}_\oplus|^2 < 0$ for all i, the subspace spanned by $\mathbf{x}_0,\ldots,\mathbf{x}_k$ and $\mathbf{f}_\oplus$ necessarily contains vectors from the interior of the cone. It follows that these Gramians are always nonpositive and zero if and only if the vectors $\mathbf{x}_i$ together with $\mathbf{f}_\oplus$ are dependent. Together with Menger's intrinsic characterization of Euclidean distances (derived in the previous section), this shows that the quantities $D(\mathbf{X}_i,\mathbf{X}_j) = |\mathbf{X}_i - \mathbf{X}_j|^2 = (\mathbf{X}_i - \mathbf{X}_j)^2$ are indeed the squares of Euclidean distances in a space of dimension at most n. Moreover, since $\mathbf{R}^{n+1,1}$ is nondegenerate and for $\mathbf{x}_i := \mathbf{e}_i - \mathbf{f}_\ominus$ $(i = 1,\ldots,n)$ the vectors $-\mathbf{f}_\ominus$, $\check{\mathbf{x}}_1,\ldots,\check{\mathbf{x}}_n \in \breve{\mathcal{N}}$ are independent by the Lemma, the dimension of the Euclidean space in which their mutual distances can be realized is in fact equal to n. $\square$

Remarks: In Klein's model of a hyperbolic space, the points are lines through the origin and inside the null cone. Viewed from this perspective, vectors on the parabola are homogenous coordinates for ideal (infinite) points in a hyperbolic space, and the parabola itself is just the set of all ideal points minus the single ideal point $\mathbf{f}_\oplus$. Alternatively, and as done by Wachter and Seidel, one can consider vectors on the parabola to be the homogeneous coordinates of a bundle of projective lines passing through a common ideal point $\mathbf{f}_{\oplus}$. In the foregoing, however, we have used Hestenes' interpretation of "points" as projective *planes* $\mathbf{X} = \mathbf{F} \wedge \mathbf{x}$. This enabled Hestenes to describe the relation between the Lorentz space $\mathbf{R}^{n+1,1}$ and conformal space $\mathbf{R}^n$ directly in terms of the conformal split, and hence to use the methods of geometric algebra to work out the homomorphism between the Lorentz group and the corresponding conformal group in detail. It will also prove useful to us in the last section of this paper.

4 Coordinates

In this section we will expand everything done in the previous section in terms of a basis. This will enable us to fill in a few missing details, and in addition it will enable us to identify the non-scalar terms of the conformal split of the geometric product of any $\mathbf{x}$, $\mathbf{y} \in \breve{\mathcal{N}}$ with specific subspaces of the vector space $C\ell_{n+1,1}$. Using the relation $\mathbf{F} = \mathbf{F}^{-1}$, we may write

$$\begin{aligned}
\mathbf{x}\,\mathbf{y} &= (\mathbf{x}\,\mathbf{F}\,\mathbf{F})(\mathbf{y}\,\mathbf{F}\,\mathbf{F}) \\
&= ((\mathbf{x}\cdot\mathbf{F} + \mathbf{x}\wedge\mathbf{F})\mathbf{F})((\mathbf{y}\cdot\mathbf{F} + \mathbf{y}\wedge\mathbf{F})\mathbf{F}) \\
&= (\mathbf{x}_{\parallel} + \mathbf{x}_{\perp})(\mathbf{y}_{\parallel} + \mathbf{y}_{\perp}) \\
&= \mathbf{x}_{\parallel}\cdot\mathbf{y}_{\parallel} \,+\, \mathbf{x}_{\perp}\cdot\mathbf{y}_{\perp} \,+\, \mathbf{x}_{\parallel}\wedge\mathbf{y}_{\parallel} \\
&\quad + \mathbf{x}_{\parallel}\wedge\mathbf{y}_{\perp} \,+\, \mathbf{x}_{\perp}\wedge\mathbf{y}_{\parallel} \,+\, \mathbf{x}_{\perp}\wedge\mathbf{y}_{\perp}
\end{aligned}$$

where $\mathbf{x}_{\parallel} := (\mathbf{x}\cdot\mathbf{F})/\mathbf{F}$ and $\mathbf{x}_{\perp} := (\mathbf{x}\wedge\mathbf{F})/\mathbf{F}$ are the components of $\mathbf{x}$ parallel and perpendicular to $\mathbf{F}$, respectively, and similarly for $\mathbf{y}_{\parallel}$ and $\mathbf{y}_{\perp}$.

If we now expand $\mathbf{x}$ and $\mathbf{y}$ in terms of the canonical basis of $\mathbf{R}^{n+1,1}$ as $\mathbf{x} = x_{-1}\mathbf{e}_{-1} + \cdots + x_n\mathbf{e}_n$ and $\mathbf{y} = y_{-1}\mathbf{e}_{-1} + \cdots + y_n\mathbf{e}_n$, we see that $\mathbf{x}\cdot\mathbf{f}_{\oplus} = \mathbf{y}\cdot\mathbf{f}_{\oplus} = -\frac{1}{2}$ implies $x_{-1} - x_0 = y_{-1} - y_0 = 1$, while $\mathbf{x}^2 = \mathbf{y}^2 = 0$ implies $x_{-1}^2 - x_0^2 = x_1^2 + \cdots + x_n^2$ and $y_{-1}^2 - y_0^2 = y_1^2 + \cdots + y_n^2$. Hence

$$\begin{aligned}
\mathbf{X}^2 &= ((x_{-1}\mathbf{e}_{-1} + \cdots + x_n\mathbf{e}_n)\wedge\mathbf{F})^2 \\
&= (x_1\mathbf{e}_1\mathbf{e}_{-1}\mathbf{e}_0 + \cdots + x_n\mathbf{e}_n\mathbf{e}_{-1}\mathbf{e}_0)^2 \\
&= x_1^2 + \cdots + x_n^2 \;=\; x_{-1}^2 - x_0^2 \\
&= (x_{-1} + x_0)(x_{-1} - x_0) \;=\; (x_{-1} + x_0)\ ,
\end{aligned}$$

i.e. $2\,\mathbf{x}\cdot\mathbf{f}_{\ominus}$.

Similarly, for the inner square

$$\begin{aligned}
|\mathbf{X}-\mathbf{Y}|^2 &= (x_1-y_1)^2+\cdots+(x_n-y_n)^2 \\
&= x_1^2+\cdots+x_n^2+y_1^2+\cdots+y_n^2 \\
&\quad -2(x_1y_1+\cdots+x_ny_n) \\
&= x_{-1}^2-x_0^2+y_{-1}^2-y_0^2 \\
&\quad -2(x_1y_1+\cdots+x_ny_n) \\
&= (x_{-1}-y_{-1})^2-(x_0-y_0)^2 \\
&\quad +2(x_{-1}y_{-1}-x_0y_0-\cdots-x_ny_n) \\
&= ((x_{-1}-y_{-1})+(x_0-y_0))((x_{-1}-y_{-1})-(x_0-y_0)) \\
&\quad +2(x_{-1}y_{-1}-x_0y_0-\cdots-x_ny_n) \\
&= 2(x_{-1}y_{-1}-x_0y_0-\cdots-x_ny_n)
\end{aligned}$$

since $\mathbf{x}\cdot\mathbf{f}_{\oplus}=\mathbf{y}\cdot\mathbf{f}_{\oplus}=-\frac{1}{2}$ implies $((x_{-1}-y_{-1})-(x_0-y_0))=0$.

Thus we have shown directly that $\mathbf{x}\cdot\mathbf{y}=\mathbf{x}_{\parallel}\cdot\mathbf{y}_{\parallel}+\mathbf{x}_{\perp}\cdot\mathbf{y}_{\perp}$ is just $-\frac{1}{2}|\mathbf{X}-\mathbf{Y}|^2$. This also shows that $[x_1,\ldots,x_n]$ are indeed the Cartesian coordinates of the "point" $\mathbf{x}$ relative to the origin $-\mathbf{f}_{\ominus}$, and hence that the set of vectors on the parabola, when equipped with the squared distance function $(\mathbf{x},\mathbf{y})\mapsto -2\,\mathbf{x}\cdot\mathbf{y}$, constitutes a complete and continuous Euclidean space $\mathbf{R}^n$. This fact could have been proven without using coordinates, but only the properties of the squared distance function itself[17].

Now let's take a look at $\mathbf{X} \triangle \mathbf{Y} := \langle \mathbf{X}\mathbf{Y} \rangle_2$:

$$\begin{aligned}
\mathbf{X} \triangle \mathbf{Y} &= \left\langle \left(\sum_{i=1}^{n} x_i \mathbf{e}_{-1}\, \mathbf{e}_0\, \mathbf{e}_i \right) \left(\sum_{j=1}^{n} y_j \mathbf{e}_{-1}\, \mathbf{e}_0\, \mathbf{e}_j \right) \right\rangle_2 \\
&= \langle \sum_{i=1}^{n} x_i y_i \,+\, \sum_{i \neq j}^{n} x_i y_j \mathbf{e}_i\, \mathbf{e}_j \,\rangle_2 \\
&= \sum_{0<i<j}^{n} (x_i y_j - x_j y_i) \mathbf{e}_i\, \mathbf{e}_j
\end{aligned}$$

In other words, $\mathbf{X} \triangle \mathbf{Y} = \mathbf{x}_\perp \wedge \mathbf{y}_\perp$ is the outer product of the Cartesian coordinate vectors. Moreover, as we shall see in the next section, $\mathbf{X} \triangle \mathbf{Y} + (\mathbf{X} - \mathbf{Y})\, \mathbf{f}_\ominus$ determines a line-bound vector in the sense of Grassmann.

For the coefficient of $\mathbf{F}$ in $\mathbf{x}_\parallel \wedge \mathbf{y}_\parallel$, we have

$$\begin{aligned}
\tfrac{1}{2}(\mathbf{Y}^2 - \mathbf{X}^2) &= \tfrac{1}{2}(y_{-1}^2 - y_0^2 - x_{-1}^2 + x_0^2) \\
&= \tfrac{1}{2}((y_{-1} + y_0) \,-\, (x_{-1} + x_0)) \\
&= \tfrac{1}{2}((x_{-1} - x_0)(y_{-1} + y_0) \,-\, (y_{-1} - y_0)(x_{-1} + x_0)) \\
&= x_{-1} y_0 - x_0 y_{-1} \ ,
\end{aligned}$$

and finally, since $\mathbf{F}\,\mathbf{f}_\oplus = \pm \mathbf{f}_\oplus$ and $1 = x_{-1} - x_0 = y_{-1} - y_0$:

$$\begin{aligned}
&(\mathbf{X}\mathbf{Y}^2 - \mathbf{Y}\mathbf{X}^2)\, \mathbf{f}_\oplus \,+\, (\mathbf{X} - \mathbf{Y})\, \mathbf{f}_\ominus \\
={}& \left(\sum_{i=1}^{n} (x_i (y_1^2 + \cdots + y_n^2) \,-\, y_i (x_1^2 + \cdots + x_n^2)) \mathbf{e}_i \right) \mathbf{f}_\oplus \\
&+ \left(\sum_{i=1}^{n} (y_i - x_i) \mathbf{e}_i \right) \mathbf{f}_\ominus \\
={}& \left(\sum_{i=1}^{n} (x_i (y_{-1} + y_0 + 1) \,-\, y_i (x_{-1} + x_0 + 1)) \mathbf{e}_i \right) \mathbf{e}_{-1}/2
\end{aligned}$$

$$
\begin{aligned}
&\quad + \left(\sum_{i=1}^{n} (x_i(y_{-1} + y_0 - 1) \;-\; y_i(x_{-1} + x_0 - 1))\mathbf{e}_i \right) \mathbf{e}_0/2 \\
&= \left(\sum_{i=1}^{n} (x_i(2y_{-1}) \;-\; y_i(2x_{-1}))\mathbf{e}_i \right) \mathbf{e}_{-1}/2 \\
&\quad + \left(\sum_{i=1}^{n} (x_i(2y_0) \;-\; y_i(2x_0))\mathbf{e}_i \right) \mathbf{e}_0/2 \\
&= \sum_{i=1}^{n} ((x_{-1}y_i - x_i y_{-1})\mathbf{e}_{-1}\,\mathbf{e}_i \;+\; (x_0 y_i - x_i y_0)\mathbf{e}_0\,\mathbf{e}_i) \ ,
\end{aligned}
$$

i.e. $\mathbf{x}_\parallel \wedge \mathbf{y}_\perp + \mathbf{x}_\perp \wedge \mathbf{y}_\parallel$.

Remarks: An interesting n-dimensional interpretation of these $(n+2)$-dimensional coordinates is as the homogenous coordinates of spheres (cf. Ref. [8]). Following Pedoe[35], we define the "inner product" of two spheres $\mathbf{s}$ and $\mathbf{s}'$ as $\mathbf{s} \cdot \mathbf{s}' = (r^2 + (r')^2 - (d(\mathbf{s}, \mathbf{s}'))^2)/2$, where r and r' are the radii of the spheres and $d(\mathbf{s}, \mathbf{s}')$ is the distance between their centers. The value of this inner product is equal to $r\,r' \cos(\theta)$, where θ is the angle in which the spheres intersect (if they intersect), and hence the inner product is zero if and only if the spheres are orthogonal. If we represent each sphere by the coordinates $\mathbf{s} = [(R+1)/2, (R-1)/2, x_1, \ldots, x_n]$, where $[x_1, \ldots, x_n]$ are the Cartesian coordinates of its center and $R = x_1^2 + \cdots + x_n^2 - r^2$, we obtain the above inner product by interpreting the sphere's coordinates as a point in $\mathbf{R}^{n+1,1}$ with respect to the basis $[\mathbf{e}_{-1}, \mathbf{e}_0, \mathbf{e}_1, \ldots, \mathbf{e}_n]$. The parabola $\breve{\mathcal{N}}$ then corresponds to the set of all spheres of zero radius[36].

5 Covariant Properties

Given a group $\mathcal{G}$ and two finite dimensional $\mathbf{R}$-linear representations $\psi : \mathcal{G} \to GL(n, \mathbf{R})$ and $\phi : \mathcal{G} \to GL(m, \mathbf{R})$, a map $\mathbf{g} : \mathbf{R}^n \to \mathbf{R}^m$ is called a *covariant* of $\mathcal{G}$ relative to ψ and ϕ if for every $\mathbf{G} \in \mathcal{G}$ the following diagram commutes:

$$\begin{tikzcd} \mathbf{R}^n \arrow[r, "\psi(\mathbf{G})"] \arrow[d, "\mathbf{g}"'] & \mathbf{R}^n \arrow[d, "\mathbf{g}"] \\ \mathbf{R}^m \arrow[r, "\phi(\mathbf{G})"] & \mathbf{R}^m \end{tikzcd}$$

In the special case that $\phi(\mathcal{G}) = \{1\}$, i.e. ϕ is the trivial representation, $\mathbf{g}$ is called an *invariant*.

If $n = n_1 + \cdots + n_k$ and $\psi = \psi_1 \times \cdots \times \psi_k$ for k (possibly identical) representations $\psi_1 : \mathcal{G} \to \mathbf{R}^{n_1}$, . . . , $\psi_k : \mathcal{G} \to \mathbf{R}^{n_k}$ of $\mathcal{G}$, the map $\mathbf{g}$ is also called a covariant of $\mathcal{G}$ relative to $\psi_1, \ldots, \psi_k$ and ϕ. In the following, $\mathcal{G}$ is usually a subgroup of $GL(n, \mathbf{R})$ and $\psi_1, \ldots, \psi_k$ all coincide with the canonical embedding of $\mathcal{G}$ into $GL(n, \mathbf{R})$. Then, the problem is to discover maps $\mathbf{g} : \mathbf{R}^n \to \mathbf{R}^m$ that are covariants of $\mathcal{G}$ with respect to some representation $\phi : \mathcal{G} \to GL(m, \mathbf{R})$. This problem is of interest because it leads one to the discovery of new representations of the group $\mathcal{G}$, for which the elements of the representation space $\mathbf{R}^m$ have a geometric and hence often (for certain groups) physical significance. Well-known examples include spinors representing fermions, and the line-bound vectors representing forces acting on as well as infinitesimal motions of rigid bodies.

If $\mathcal{G}$ denotes the group of isometries of $\mathbf{R}^{p,q}$, it acts naturally on the corresponding Clifford algebra $C\ell_{p,q}$, defining a representation $\phi : \mathcal{G} \to GL(2^{p+q}, \mathbf{R})$ such that $\phi(\mathbf{G})(\mathbf{v}) = \mathbf{G}(\mathbf{v})$ for every $\mathbf{G} \in \mathcal{G}$ and $\mathbf{v} \in \mathbf{R}^{p+q}$. The geometric product of vectors is a covariant (mapping $\mathbf{R}^{p,q} \times \cdots \times \mathbf{R}^{p,q}$ to $C\ell_{p,q}$) of the group of isometries of $\mathbf{R}^{p,q}$ with respect to its natural action on $C\ell_{p,q}$. These claims may be proven by recalling that the isometries can be written by conjugation with a product of unit vectors $\mathbf{P} \in \mathbf{Pin}(p,q)$, i.e. $\mathbf{x} \mapsto \mathbf{P}\,\mathbf{x}\,\overline{\mathbf{P}}$, while for any $\mathbf{x}_1, \ldots, \mathbf{x}_k \in \mathbf{R}^{p,q}$ associativity together with $\overline{\mathbf{P}}\,\mathbf{P} = \pm 1$ implies

$$\mathbf{P}\,\mathbf{x}_1\,\overline{\mathbf{P}} \cdots \mathbf{P}\,\mathbf{x}_k\,\overline{\mathbf{P}} \;=\; (\overline{\mathbf{P}}\,\mathbf{P})^{k-1}\,\mathbf{P}\,\mathbf{x}_1 \cdots \mathbf{x}_k\,\overline{\mathbf{P}}\,.$$

This result extends easily to products of vectors and thence, by linearity, to arbitrary multivectors in $C\ell_{p,q}$, thus showing that the geometric product of arbitrary multivectors is actually a covariant of the group of isometries with respect to its natural action on $C\ell_{p,q}$. We shall refer to the representation defined on the products of vectors by $\mathbf{x}_1 \cdots \mathbf{x}_k \mapsto (\mathbf{P}\,\overline{\mathbf{P}})^{k-1}\,\mathbf{P}\,\mathbf{x}_1 \cdots \mathbf{x}_k\,\overline{\mathbf{P}}$ as the *extension* of the isometries to $C\ell_{p,q}$.[37]

By virtue of its canonical embedding in $C\ell_{p,q}$, the isometries also act in a natural way on the space $\bigwedge_k(\mathbf{R}^{p,q})$ generated by all k-vectors, and the projection operator $\langle\rangle_k : C\ell_{p,q} \to \bigwedge_k(\mathbf{R}^{p,q})$ is easily seen to be a covariant relative to this group action. Note further that given any three group representations $\psi_i : \mathcal{G} \to GL(n_i, \mathbf{R})$, $i = 1, 2, 3$, the composition of a covariant $\mathbf{g}_{12} : \mathbf{R}^{n_1} \to \mathbf{R}^{n_2}$ with respect to ψ_1, ψ_2 and a covariant $\mathbf{g}_{23} : \mathbf{R}^{n_2} \to \mathbf{R}^{n_3}$ with respect to ψ_2, ψ_3 is a covariant $\mathbf{g}_{23} \circ \mathbf{g}_{12}$ with respect to ψ_1, ψ_3. Because the inner and outer products of a k-vector with an m-vector are the $|k-m|$-vector and $(k+m)$-vector parts of their geometric product, respectively, it follows that the inner and outer products are also covariant operations, once again with respect to the natural group actions. The inner product of two k-vectors is in fact an invariant!

These considerations show that we can use Clifford algebras to construct not only interesting families of representations of isometry groups, but also a large family of covariants with respect to these representations. They also show that Clifford algebra can greatly facilitate the calculations involved: If we had used matrices to represent the isometries, then to prove the covariance of these products it would have been necessary to construct the representation $\phi : \mathcal{G} \to GL(m, \mathbf{R})$ in terms of matrices. Since we have represented the transformations by conjugation with respect to elements of $\mathbf{Pin}(p, q)$, however, the new group action is also represented by conjugation by the same elements (up to a factor of ± 1) and is obviously a representation. The check for commutativity was also relatively easy. It is now straightforward to show that given any subgroup $\mathcal{G} \subseteq \mathbf{Pin}(p, q)$, a subspace $\mathcal{U}$ of $\mathbf{R}^{p,q}$ that is preserved by $\mathcal{G}$, and a basis $\mathbf{u}_1, \ldots, \mathbf{u}_k$ of $\mathcal{U}$, both the inner and outer products of any multivector with $\mathbf{U} := \mathbf{u}_1 \wedge \cdots \wedge \mathbf{u}_k$ are covariants (even if we do not let $\mathcal{G}$ act on $\mathbf{U}$!). Since orthogonal projection onto $\mathcal{U}$ can be written as $\mathbf{x} \mapsto (\mathbf{x} \cdot \mathbf{U})\mathbf{U}^{-1}$ (assuming that $\mathbf{U}^{-1}$ exists), this in fact shows that such projections are likewise covariants.

Hestenes has used the fact that the group of isometries of $\mathbf{R}^{p+1,q+1}$, when composed with the rescaling necessary to keep $\mathbf{x} \cdot \mathbf{f}_{\oplus} = -\frac{1}{2}$, corresponds under the conformal split to the group of all conformal transformations of $\mathbf{R}^{p,q}$ [1] (note that the vectors $\mathbf{f}_{\oplus}$ are *not* transformed!). He has also shown (in effect) that the conformal split is covariant, at least up to the scale factor needed to maintain $\mathbf{x} \cdot \mathbf{f}_{\oplus} = -\frac{1}{2}$. To see this, let $\mathbf{g} : \breve{\mathcal{N}} \to C\ell_{p,q}$ be defined by $\mathbf{x} \mapsto \mathbf{F}\,\mathbf{x}$, and let $\phi : \mathbf{Pin}(p+1, q+1) \to \mathbf{Fpin}(p+1, q+1)$ be the involutory isomorphism defined by $\mathbf{G_P}(\mathbf{x}) := \mathbf{P}\,\mathbf{x}\,\overline{\mathbf{P}} \mapsto \phi(\mathbf{G_P})(\mathbf{x}) := \mathbf{F}\,\mathbf{P}\,\mathbf{F}\,\mathbf{x}\,\overline{\mathbf{P}}$, where $\mathbf{Fpin}(p+1, q+1) := \phi(\mathbf{Pin}(p+1, q+1))$. We may then check the commutativity of our diagram:

$$\phi(\mathbf{G_P})(\mathbf{g}(\mathbf{x})) = \mathbf{F}\,\mathbf{P}\,\mathbf{F}\,\mathbf{F}\,\mathbf{x}\,\overline{\mathbf{P}} = \mathbf{F}\,\mathbf{P}\,\mathbf{x}\,\overline{\mathbf{P}} = \mathbf{g}(\mathbf{P}\,\mathbf{x}\,\overline{\mathbf{P}}) = \mathbf{g}(\mathbf{G_P}(\mathbf{x}))\ .$$

When $\mathbf{G_P}(\mathbf{x})$ is divided its inner product with $-2\,\mathbf{f}_{\oplus}$, however, $\mathbf{g}$ will no longer be covariant in general. Fortunately, the subgroup of the conformal group generated by translations and rotations corresponds to the stabilizer of $\mathbf{f}_\oplus$ under the conformal split, so that this inner product is always unity in this subgroup.

These further considerations suggest that Hestenes' conformal split will be useful in the study of Euclidean covariants (i.e. the case that $p = n$ and $q = 0$). We have in fact already seen how it can be used to show that the fundamental invariant of Euclidean geometry, namely the squared distance, is equal to the inner product of vectors on the parabola, and more generally that for $\mathbf{x}_1, \ldots, \mathbf{x}_k \in \breve{\mathcal{N}}$ the inner square $|\mathbf{x}_1 \wedge \cdots \wedge \mathbf{x}_k \wedge \mathbf{f}_\oplus|^2$ is essentially just a Cayley-Menger determinant, which is proportional to the squared volume of the simplex spanned by the "points" $\mathbf{X}_i := \mathbf{F} \wedge \mathbf{x}_i$. The above considerations also show that the outer product of $\mathbf{x}, \mathbf{y} \in \breve{\mathcal{N}}$ is likewise a Euclidean covariant, namely $\mathbf{x} \wedge \mathbf{y} = \mathbf{X} \triangle \mathbf{Y} + \frac{1}{2}(\mathbf{Y}^2 - \mathbf{X}^2)\mathbf{F} + \mathbf{X}(\mathbf{Y} - \mathbf{X})\mathbf{Y}\, \mathbf{f}_\oplus + (\mathbf{X} - \mathbf{Y})\, \mathbf{f}_\ominus$. As we shall see momentarily, however, this contains several simpler covariants.

Let us illustrate these ideas by considering the action of the subgroup $\mathcal{G} := \{\mathbf{S} \in \mathbf{Spin}(n+1,1) \mid \mathbf{S} \cdot \mathbf{F} = 0\}$ on the product of two vectors on the parabola $\mathbf{x}, \mathbf{y} \in \breve{\mathcal{N}}$, and its geometric interpretation in the Euclidean geometry obtained via the conformal split. As explained in Hestenes[1], this subgroup corresponds to the rotations about the origin in the Euclidean space, and it preserves the vectors $\mathbf{f}_\oplus$. Hence $(\mathbf{x} \wedge \mathbf{y}) \cdot \mathbf{F} = \frac{1}{2}\,(\mathbf{Y}^2 - \mathbf{X}^2)$ is a covariant (actually an invariant!), as is $(((\mathbf{x} \wedge \mathbf{y}) \cdot \mathbf{f}_\ominus)\, \mathbf{f}_\ominus) \wedge \mathbf{f}_\oplus = \frac{1}{4}\mathbf{X}(\mathbf{Y} - \mathbf{X})\mathbf{Y}$ and $(((\mathbf{x} \wedge \mathbf{y}) \cdot \mathbf{f}_\oplus)\, \mathbf{f}_\oplus) \wedge \mathbf{f}_\ominus = \frac{1}{4}(\mathbf{X} - \mathbf{Y})$. On the other hand, since $\mathbf{F}^\perp$ is also preserved, taking the projection of $\mathbf{x} \wedge \mathbf{y}$ onto $\mathbf{e}_1 \wedge \cdots \wedge \mathbf{e}_n$ shows that $\mathbf{X} \triangle \mathbf{Y}$ is a covariant.

To see how these quantities actually transform, take $\mathbf{S} \in \mathbf{Spin}(n+1,1)$ with $\mathbf{S} \cdot \mathbf{F} = 0$ and compute:

$$\mathbf{S}\,\mathbf{x}\,\tilde{\mathbf{S}}\,\mathbf{S}\,\mathbf{y}\,\tilde{\mathbf{S}} \quad = \quad \mathbf{S}\,\mathbf{x}\,\mathbf{F}\,\mathbf{F}\,\mathbf{y}\,\tilde{\mathbf{S}}$$

$$
\begin{aligned}
&= \mathbf{S}(-\mathbf{f}_\ominus - \mathbf{X}^2\mathbf{f}_\oplus + \mathbf{X})(\mathbf{f}_\ominus + \mathbf{Y}^2\mathbf{f}_\oplus + \mathbf{Y})\tilde{\mathbf{S}} \\
&= -\tfrac{1}{2}(|\mathbf{X}-\mathbf{Y}|^2 + (\mathbf{X}^2 - \mathbf{Y}^2)\mathbf{F}) + \mathbf{S}\,(\mathbf{X}\triangle\mathbf{Y})\,\tilde{\mathbf{S}} \\
&\quad +\mathbf{S}\,\mathbf{X}(\mathbf{Y}-\mathbf{X})\mathbf{Y}\,\tilde{\mathbf{S}}\,\mathbf{f}_\oplus + \mathbf{S}(\mathbf{X}-\mathbf{Y})\tilde{\mathbf{S}}\,\mathbf{f}_\ominus\,.
\end{aligned}
$$

From this we see that $\mathbf{X}-\mathbf{Y}$ and $\mathbf{X}(\mathbf{Y}-\mathbf{X})\mathbf{Y}$ are relatively trivial covariants of the rotation group, since they transform the same way that the Cartesian coordinates $\mathbf{X},\mathbf{Y}$ do, but $\mathbf{X}\triangle\mathbf{Y}$ transforms with the extension of the rotation to $C\ell_n$.

We now consider the action of the group of "translators" $\mathbf{T_v} := \{1 + \mathbf{v}\,\mathbf{f}_\oplus \mid \mathbf{v} \in \mathbf{R}^{n+1,1}, \mathbf{v}\cdot\mathbf{F} = 0\}$ on the bivector $\mathbf{x}\wedge\mathbf{y}$. As shown by Hestenes, under the conformal split these correspond to translations $\mathbf{X}\mapsto\mathbf{X}+\mathbf{V}$, satisfy $\tilde{\mathbf{T}}_\mathbf{v} = 1-\mathbf{v}\,\mathbf{f}_\oplus = \mathbf{F}\,\mathbf{T_v}\,\mathbf{F}$, and preserve only the "point at infinity" $\mathbf{f}_\oplus$. Hence $\mathbf{x}\wedge\mathbf{y}\wedge\mathbf{f}_\oplus = (\mathbf{X}\triangle\mathbf{Y})\,\mathbf{f}_\oplus + \frac{1}{2}(\mathbf{X}-\mathbf{Y})$ is a covariant of *both* the groups of translations and rotations, and hence of their semidirect product, the group of Euclidean motions[38]. This provides us with an interpretation of Grassmann's line-bound vector in the context of the model for Euclidean geometry we have used in this paper; interestingly, its inner square can also be viewed as a Cayley-Menger determinant.

On the other hand, $(\mathbf{x}\wedge\mathbf{y})\cdot\mathbf{f}_\oplus = \frac{1}{2}(\mathbf{Y}^2-\mathbf{X}^2)\mathbf{f}_\oplus + \frac{1}{2}(\mathbf{Y}-\mathbf{X})\mathbf{F}$ is a covariant that has not, to our knowledge, previously been considered. To see how the components transform, we compute:

$$
\begin{aligned}
\mathbf{T_v}\,(\mathbf{x}\,\mathbf{y})\,\tilde{\mathbf{T}}_\mathbf{v} &= (\mathbf{F}\mathbf{F})(\mathbf{T_v}\,\mathbf{x}\,\tilde{\mathbf{T}}_\mathbf{v})(\mathbf{F}\mathbf{F})(\mathbf{T_v}\,\mathbf{y}\,\tilde{\mathbf{T}}_\mathbf{v}) \\
&= (\mathbf{F}\,\tilde{\mathbf{T}}_\mathbf{v}\,(\mathbf{F}\,\mathbf{x})\,\tilde{\mathbf{T}}_\mathbf{v}\,\mathbf{F})\,(\tilde{\mathbf{T}}_\mathbf{v}(\mathbf{F}\,\mathbf{y})\tilde{\mathbf{T}}_\mathbf{v}) \\
&= (-\mathbf{f}_\ominus - (\mathbf{X}-\mathbf{V})^2\mathbf{f}_\oplus + \mathbf{X}-\mathbf{V})
\end{aligned}
$$

$$(\mathbf{f}_\ominus + (\mathbf{Y}-\mathbf{V})^2\mathbf{f}_\oplus + \mathbf{Y}-\mathbf{V})\ ,$$

where $\mathbf{V} := \mathbf{F}\,\mathbf{v}$. Here, we have used the fact that conjugation of $\mathbf{F}\,\mathbf{x}$ by $\mathbf{F}$ corresponds to the inversion in the origin $\mathbf{X} \mapsto -\mathbf{X}$ and rescaling of $\mathbf{F}\,\mathbf{x}$ by -1 (see equation (5.71) in Hestenes[1]). It follows that

$$\begin{aligned}\mathbf{T_v}\,(\mathbf{x}\,\mathbf{y})\,\tilde{\mathbf{T}}_\mathbf{v} \;&=\; \mathbf{x}\,\mathbf{y} + \mathbf{V}\triangle(\mathbf{Y}-\mathbf{X}) + (\mathbf{V}\cdot(\mathbf{Y}-\mathbf{X}))\,\mathbf{F}\\ &\quad + (\mathbf{X}-\mathbf{V})(\mathbf{Y}-\mathbf{X})(\mathbf{Y}-\mathbf{V})\,\mathbf{f}_\oplus\\ &\quad - \mathbf{X}(\mathbf{Y}-\mathbf{X})\mathbf{Y}\,\mathbf{f}_\oplus\ .\end{aligned}$$

Thus, under translation by $\mathbf{V}$ the vector $\mathbf{Y}-\mathbf{X}$ contributes to the components of $\mathbf{X}\triangle\mathbf{Y}$ as $\mathbf{V}\triangle(\mathbf{Y}-\mathbf{X})$, and to the component $\frac{1}{2}(\mathbf{Y}^2-\mathbf{X}^2)\mathbf{F}$ as $(\mathbf{V}\cdot(\mathbf{Y}-\mathbf{X}))\mathbf{F}$.

Remarks: Clearly, the above preliminary study could be extended to the products of three and more vectors $\mathbf{x},\mathbf{y},\mathbf{z} \in \breve{\mathcal{N}}$, thus yielding the joint covariants of three and more points. We believe that many heretofore unknown covariants, and possibly even some interesting invariants, of the Euclidean group remain to be discovered in this way. For example, it is easily seen that the oriented volume of the simplex spanned by $n+1$ points is equal to $\frac{1}{n!}$ times the coefficient of $\mathbf{f}_\ominus$ in the product of the corresponding vectors on the parabola. Other, less trivial examples will probably be found in the invariants of two Euclidean subspaces (antisymmetric Euclidean tensors) that were first characterized by J. J. Seidel[18]. It may even be possible to use this approach to prove the first and second fundamental theorems of invariant theory for the Euclidean group[21].

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References

[1] D. Hestenes. The design of linear algebra and geometry. *Acta Appl. Math.*, 23:65–93, 1991.

[2] H. A. Kastrup. Zur physikalischen Deutung und darstellungstheoretischen Analyse der konformen Transformationen von Raum und Zeit. *Ann. Phys.*, 9:388–428, 1962.

[3] A. Crumeyrolle. *Orthogonal and Symplectic Clifford Algebras, Spinor Structures*. Kluwer Acad. Publ., Dordrecht, 1990.

[4] D. Hestenes. Universal geometric algebra. *Simon Stevin*, 62:253–274, 1988.

[5] L. V. Ahlfors. Möbius transformations and Clifford numbers. In I. Chavel and H. M. Farkas, editors, *Differential Geometry and Complex Analysis*. Springer-Verlag, Berlin, 1985.

[6] J. Maks. *Modulo (1,1) Periodicity of Clifford Algebras and Generalized (anti-)Möbius Transformations*. PhD thesis, T.U. Delft, 1989.

[7] P. Lounesto and A. Springer. Möbius transformations and Clifford algebras of euclidean and anti-euclidean spaces. In J. Lawrynowicz, editor, *De-*

formations of Mathematical Structures, pages 79–90. Kluwer Acad. Publ., Dordrecht, 1989.

[8] J. P. Fillmore and A. Springer. Möbius groups over general fields using Clifford algebras associated with spheres. *Internat. J. Theoret. Phys.*, 29:225–246, 1990.

[9] F. Klein. *Elementary Mathematics from an Advanced Standpoint: Geometry.* Dover Publications, Inc., London, England, 1939. Translated from the original in German.

[10] M. J. Crowe. *A History of Vector Analysis.* Univ. of Notre Dame Press, 1967.

[11] D. Hestenes. *New Foundations for Classical Mechanics.* D. Reidel Publ. Co., Dordrecht, NL, 1986.

[12] J. A. Thorpe. *Elementary Topics in Differential Geometry.* Springer-Verlag, New York, NY, 1979.

[13] K. Menger. Untersuchungen über allgemeine Metrik. *Math. Annal.*, 100:75–163, 1928.

[14] K. Menger. New foundation of euclidean geometry. *Amer. J. Math.*, 53:721–745, 1931.

[15] J. J. Seidel. Distance-geometric development of two-dimensional euclidean, hyperbolic and spherical geometry. *Simon Stevin*, 29:32–50, 65–76, 1952.

[16] L. M. Blumenthal. *Theory and Applications of Distance Geometry.* Cambridge Univ. Press, Cambridge, U.K., 1953. Reprinted by the Chelsea Publishing Co., 1970.

[17] L. M. Blumenthal. *A Modern View of Geometry.* Dover Publications, Ltd., 1961.

[18] J. J. Seidel. Angles and distances in n-dimensional euclidean and non-euclidean geometry. I–III. *Indag. Math.*, 17:329–340,535–541, 1955. Reprinted *Nederl. Akad. Wetensch. Proc.*.

[19] J. Dieudonné and J. B. Carrell. Invariant theory, old and new. *Adv. Math.*, 4:1–80, 1970. Reprinted by Academic Press, 1971.

[20] H. Kraft. *Geometrishe Methoden in der Invariantentheorie.* Friedr. Vieweg & Sohn mbH, Braunschweig, Germany, 1985.

[21] D. R. Richman. The fundamental theorems of vector invariants. *Adv. Math.*, 73:43–78, 1989.

[22] D. Hilbert. Ueber die vollen Invariantensysteme. *Math. Ann.*, 42:313–373, 1893.

[23] Ch. S. Fisher. The death of a mathematical theory: A study in the sociology of knowledge. *Arch. History Exact Sci.*, 3:137–159, 1966.

[24] D. Hestenes and R. Ziegler. Projective geometry with Clifford algebra. *Acta Appl. Math.*, 23:25–63, 1991.

[25] N. L. White. Invariant-theoretic computation in projective geometry. In *DIMACS Series in Discrete Mathematics and Theoretical Computer Science*, volume 6, pages 363–377. Amer. Math. Soc., 1991.

[26] N. L. White. Multilinear Cayley factorization. *J. Symb. Comput.*, 11:421–438, 1991.

[27] T. McMillan and N. L. White. The dotted straightening algorithm. *J. Symb. Comput.*, 11:471–482, 1991.

[28] B. Buchberger. History and basic features of the critical-pair/completion procedure. *J. Symb. Comput.*, 3:3–38, 1987.

[29] B. Sturmfels and N. White. Gröbner bases and invariant theory. *Adv. Math.*, 76:245–259, 1989.

[30] T. F. Havel. Some examples of the use of distances as coordinates for euclidean geometry. *J. Symbolic Comput.*, 11:579–593, 1991.

[31] Dalbec, J. and Sturmfels, B. Personal communication.

[32] A. W. M. Dress and T. F. Havel. Fundamentals of the distance geometry approach to the problems of molecular conformation. In *Proc. INRIA Workshop on Computer-Aided Geometric Reasoning*, 1987.

[33] G. M. Crippen and T. F. Havel. *Distance Geometry and Molecular Conformation*. Research Studies Press, Letchworth, U.K., 1988. Publisher in the U.S. is J. Wiley & Sons, New York, NY.

[34] Hestenes actually defines this using right-multiplication by **F**.

[35] D. Pedoe. *A Course of Geometry for Colleges and Universities*. Cambridge Univ. Press, Cambridge, England, 1970.

[36] This coordinatization of the space of spheres seems to have been invented by H. Grassmann, in order to demonstrate that the coefficients of equations can be viewed as coordinates for their loci. It was further developed by W. Clifford, S. Lie, E. Müller and E. Study. In particular, in a paper in the Proceedings of the London Mathematical Society in 1868, entitled “On the

Powers of Spheres", W. Clifford showed that it yields a representation of Cayley-Menger determinants that is fully equivalent to that subsequently found by J. Seidel (*loc cit.*). Clifford later expanded upon this work in an apparently unpublished manuscript entitled "Of Power Coordinates in General". The equivalence of Clifford's results to Seidel's, however, is not obvious from casual reading, and few people in modern times seem to be familiar Clifford's early work in this area (see J. L. Coolidge, "A History of Geometrical Methods", Oxford Univ. Press, 1940, pp. 166–170).

[37] Hestenes prefers the term *outermorphism*.

[38] E. Snapper and R. J. Troyer. *Metric Affine Geometry.* Academic Press, 1971. Reprinted by Dover Publications, 1989.